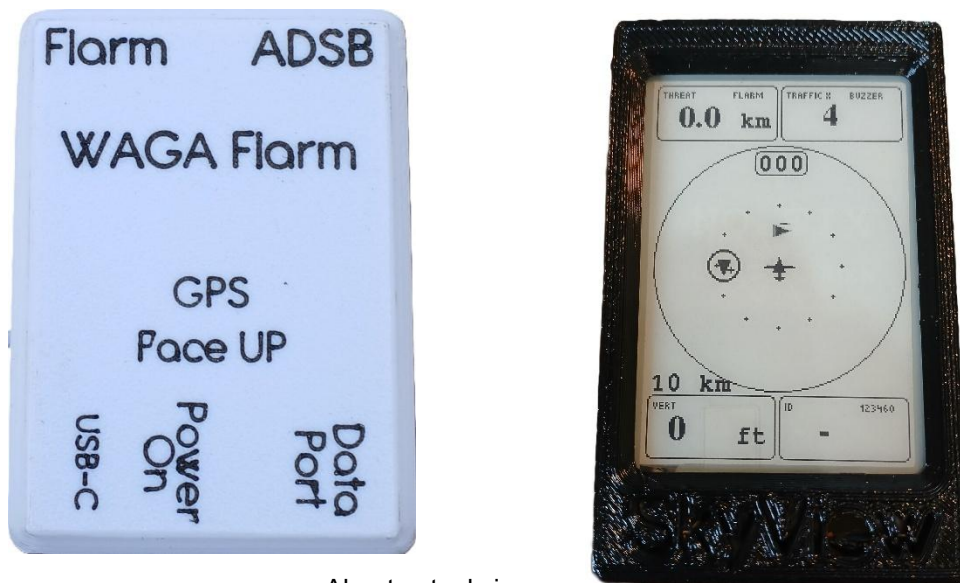


WAGA Flarm



About actual size

A fully compatible PowerFlarm with ADSB-In.
Produced by WAGA to provide a cost effective PowerFlarm.

Quick Start Manual

WAGA Flarm

QUICK START MANUAL

Please read this manual carefully before using the device. This manual is an essential component of the device and should be kept in a safe place so that it may be referred to when required.

This manual is specific to the version 1.0.A04 of the WAGA Flarm.

Never fly with a WAGA Flarm without extensive familiarisation on ground. Train on the ground, before flying, on how to interpret and react in case of a Flarm Alarm warning. Observe restrictions and safety instructions detailed in this manual and your display manual.

WARNING AND CONDITIONS OF USE

The pilot is ultimately responsible for all flight decisions and for operating the aircraft safely at all times.

Pilots must maintain good visual lookout to avoid collision.

FLARM cannot warn reliably in all situations. FLARM does not issue resolution advisories. FLARM can only warn of aircraft which are equipped with FLARM (or transponders and ADS-B Out equipment with specific FLARM devices) and of obstacles stored in its database. The use of FLARM does not allow a change of flight tactics or pilot behaviour. It is the sole responsibility of the pilot in command to decide upon the use of FLARM. FLARM Technology Ltd and WAGA cannot be held liable under any circumstances.

For situational awareness only. Never make safety critical decisions based on displayed information.

No liability is accepted by the producers of the WAGA Flarm, the displays, or its software.

Refer to the manuals on the displays you connect to the device.

The WAGA Flarm is an Atom FLARM chip compatible device (not PowerFlarm and not Classic which are older devices). It works on data port protocol version 9 and possibly higher with future updates. But is backwards compatible to previous versions.

WF is supplied with the latest FLARM firmware update Version 7.40 (April 2025) which has disabled firmware expiration, ie it will no longer stop working after 2 years. We recommend to update every FLARM device once every 12 months. Regular updates ensure the product benefits from bug fixes and new features. We recommend to include the update schedule in your regular maintenance and documentation.

The objective has been to supply you with a kit of parts customized to your aircraft. Ie your kit will provide you what you ordered with cables or we will help customize your cables if not standard. It has also been set with your aircraft details and the latest firmware so it should work as is for a year. You only need to fiddle with updates thereafter and we intend to make that easy enough.

A lot of this manual is therefore for info in case you need it. You mainly need to get to know the Display manual.

Terminology used in this manual:

CARP	Continuous Analyzer of Radio Performance. The device continuously records range information during operation when new aircraft appear and disappear. This data is aggregated into RF range statistics that can be read by data port consumers. It provides stats on reception strength for testing.
EC	Electronic Conspicuity. An aviation electronic device that transmits and receives radio signals with the intention of improving situational awareness and collision avoidance. Eg Flarm, ADSB, Mode-S, SoftRF. Comprises a transceiver and a display, which may be integrated but are more usually separate components.
FLARM©	An aviation EC system. The registered trademark of FLARM© Technology AG.
Flarm Alarms	Alarm levels assessed by Flarm EC devices giving a warning of time to impact. Defined by FLARM.
NMEA	National Marine Electronics Association standard protocol for data encoding. Flarm work on NMEA protocol but some devices also work GDL 90 which is a Garmin GA protocol not natively compatible. The only reason to know about both is that all your devices must be configured; to talk one or the other, at the correct Baud, on the correct Port. This can be a common simple problem at initial setup.
PowerFlarm	The processor chip set with FLARM© propriety software which is built into different manufacturers EC devices. Successor to the 'Classic' FLARM chip set.
SoftRF	A "DIY, multi-functional, compatible, sub-1 GHz ISM band radio-based proximity awareness system for general aviation". Open-source software, not Flarm but works with it to some extent.
SkyView	An aviation electronic conspicuity display from the SoftRF open-source project. It is Flarm compatible.
SWD	SkyView WAGA Display. A development of the SoftRF SkyView EZ.
ThisAircraft	The aircraft in which the SWD is installed
Threat	The traffic with the highest level of Flarm Alarm or, if no Flarm Alarms, then the closest traffic.
Traffic	Aircraft detected and available for display on the SWD
UART	A serial communication protocol. TTL or RS232 are other versions that are similar but not compatible. The protocol sets voltage levels and requirements.
WAGA	Western Australia Gliding Association (WAGA) Inc. Producers of the SWD and WAGA Flarm.
WAGA Flarm	An EC transceiver based on the PowerFlarm Atom. Produced not-for-profit.
WF	The WAGA Flarm device
WebUI	Web User Interface, the application used on a smart phone to configure the SWD. We may extend this to configure the WAGA Flarm and even do updates.

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1 Features

WAGA Flarm (WF) is a modern device designed using the latest FLARM© Atom chipset and the ADSB-In module to provide a completely PowerFlarm compatible device. It is the same as all compatible Flarm devices that use the FLARM© chips and firmware under license and should be updated using the Flarm Atom firmware updates provided by FLARM©. It is based on the same hardware as the latest device from Flarm, the PowerFlarm Flex. But it was designed as a simpler, low cost, no bells and whistles device to encourage every glider in WA to afford to have a Flarm and see GA traffic that is equipped with transponders or ADSB-Out. It was assembled in WA and is fully serviced in WA by WAGA members.

WAGA Flarm is for gliders and tugs in WA at this stage but could be used in any aircraft or worldwide as it has the latest worldwide capabilities. It transmits at 20dbm, the highest output allowed and has been received and receives at 50km. It is compatible with all FLARM© certified screens and glide computers. It can provide output to XCSOAR and other software in the NMEA standard protocol for GPS and Flarm alerts and alarms.

The primary function is to provide advanced glider collision avoidance features and ADSB-In situational awareness. It promotes lookout by not attracting attention in the cockpit and providing clear alarms by buzzer (or Voice) and visual indicators on the SkyView screen. It avoids extra alerts for tugs in high traffic or while on tow. It will work compatible full feature colour screens the same as any Flarm.

It is paired with the SkyView display which is a low power, large sunlight readable, clear black and white display. It is absolutely clear in sunlight. It can have a flip up (or removable) cover to protect the screen from sun as no screen lasts well in the sun. The ePaper screen works in sun but is best protected when not in use. It is low cost and easily repaired at low cost. It also provides many functions such as WiFi, Bluetooth, SD card, buzzer, voice alert output to speaker or headsets, and is programmed by WAGA and can be advanced by software updates if we wish.

Multiple power options. USB-C or 8 to 36V aircraft power in. It outputs 5V or 3.3V for displays. It can have a Li-Ion battery with advanced BMS for protection to make it a portable device that provides backup functionality or hours of standalone operation. There are complexities in having a battery which are not warranted in normal cases, but it can be added.

It uses remote mounted antenna for Flarm and ADSB so the installation can be optimised. The most critical part of installation.

It is fully testable using another WAGA Flarm (WF) and a computer, OGN, and CARP in IGC files. It can be analysed by trained WAGA members and repaired by WAGA.

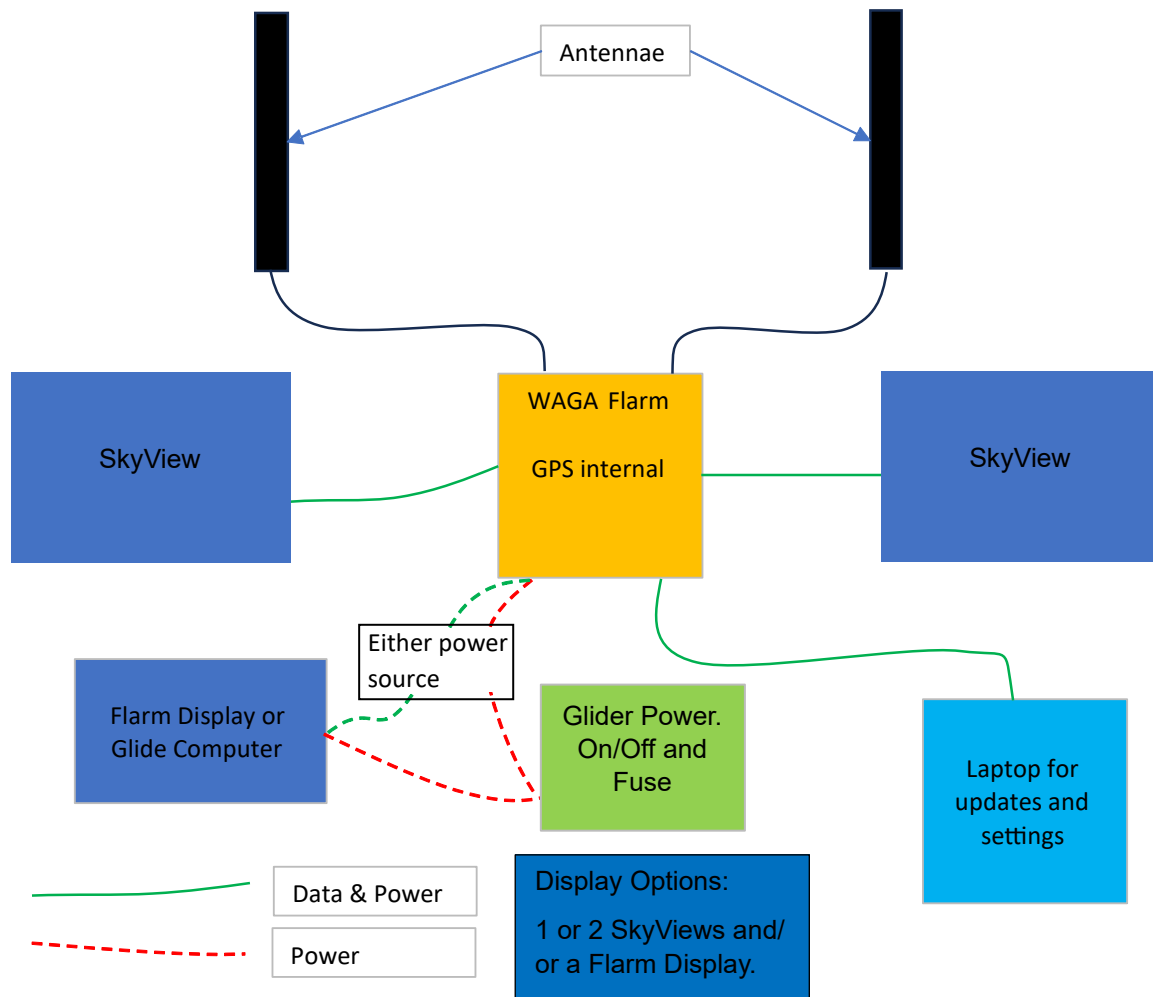
It is produced at cost by volunteers and will be installed, tested and serviced by trained WAGA members. We designed it and developed it; we can fix it at low cost.

Other Flarms can provide more features like dual Flarm Antennae or WiFi setup and Bluetooth comms. Consider other Flarms if you need or want these features.

This manual outlines essential setup and operational steps to get started quickly. For detailed instructions refer to the full manual and to manuals from Flarm. Most of the FLARM© manuals are useful but there are slight variations due to hardware type.

The WAGA Flarm kits are generally supplied with everything you need and, in most cases, already assembled with cables and the latest firmware and database registrations to save the user dealing with a lot of technical setup. Later you will need to update or change settings but initially install it in the glider and it should work. Some installs need to be customized by cable lengths and install locations. We have not supplied ready-made cables for these aircraft (tugs, motorgliders and dual seaters) but included the needed plugs. This manual gives details on making your own cables if required.

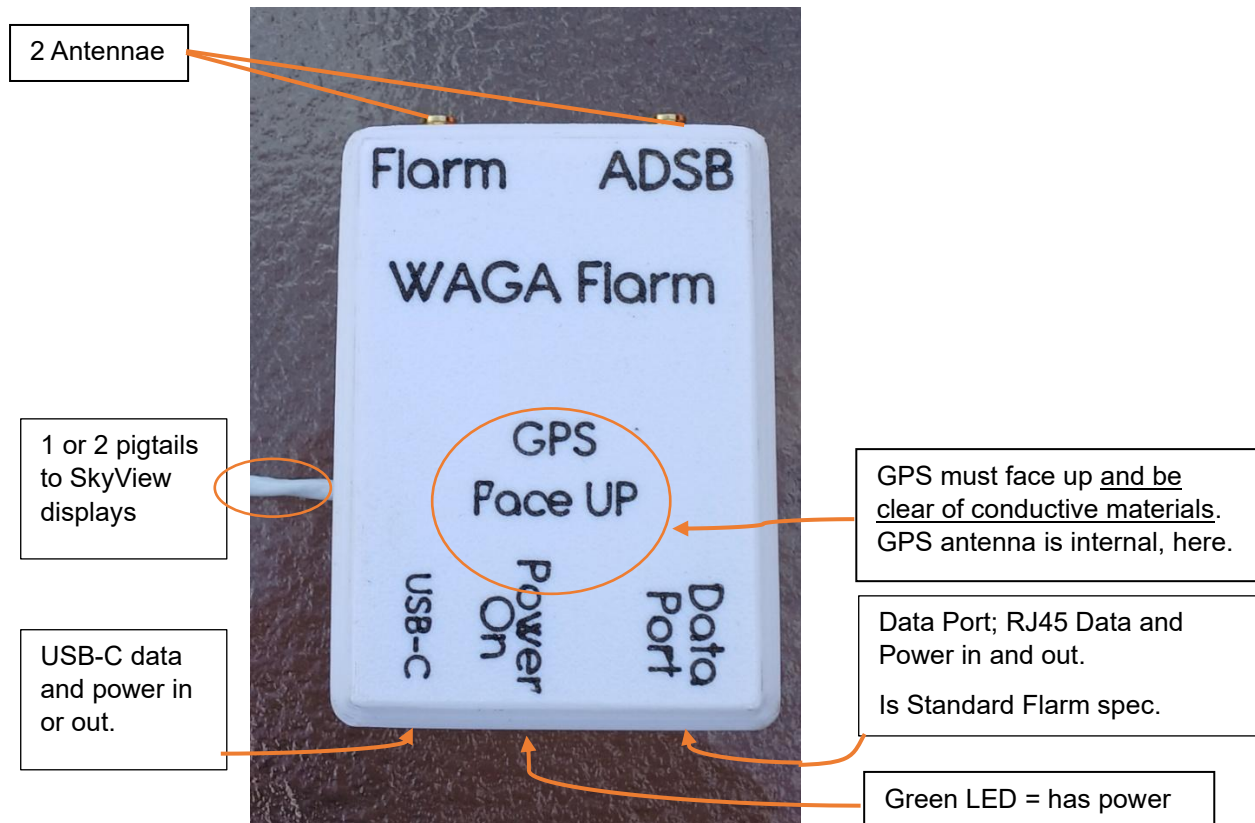
2 Installation



You must select and install some of the above components in the glider. Details are given below about which components you may select and the cables that connect them. The WAGA Flarm (WF) needs to be supplied power, connected to displays and antennae, and connected via USB to a computer when needed for updates and settings. It can also supply power. There are options and not all are required.

2.1 Connectors

This section shows the various connections to the WF:



1. **Data Port;** An RJ45 standard Flarm connector which is ICAO compatible. This should connect to any other Flarm compatible device, display, or glide computer. It can provide DC power to the unit (8 to 36V), or 3.3V and will send and receive standard serial data in the RS232 standard and ICAO and Flarm NMEA protocol. It can provide 3.3V power to displays. It is critical that the connections and polarities are correct, damage is likely if not. The NMEA protocol is an extensive standard set of sentences of data that are either sent about once per second such as GPS position and barometer altitude and alarms as required, or alerts, or status when requested. A suitable display can request CARP and IGC file output if it is capable. It cannot be used for updates (these are performed via the USB). The port must be configured to match the displays for Baud rate (19,200 is standard) and the sentences transmitted or it will not talk to the display. The Flarm settings can be changed via this port but more commonly via the USB port.
2. **USB-C connector;** This is a bi-directional USB-C Type 2. It can supply or receive 5V power. It uses a CP2102 USB chip. The Virtual COM Port (VCP) drivers must be installed on the host computer to use this feature. In Windows, use the "Device Manager" to see which COM-port is used. It should plug and play. The baud rate to match the software connected must be set on the unit, standard is 57,600bps. It sends and receives all the same data, or change settings, as the Data Port but can also be used to update firmware.
3. **UART to SkyView display.** This is provided by pigtails soldered onto the WF. There can be two for a front and rear SkyView display. They provide the same data as the Data Port but in TTL or UART standard. Usually and currently the SkyView only receives data and the connection can be a 3-wire cable to provide ground, power and data out. A 4-wire cable is supplied for the front SkyView and for the possibility that in future we will add controls to the SkyView. Only 1 of the SkyViews or the RJ45 device may provide commands and it is best to ensure this by leaving the data-in (RXD in the WF) disconnected or deactivated.
4. Two displays can be connected at a time, rather don't connect 3 displays or devices although more may work at your risk.
5. Two antennae SMA connectors are provided. One is for the ADSB and the other for the Flarm radios. The Flarm is bidirectional, the ADSB is receive only. The folded flat dipole antenna can be connected to either. If you use the Flarm dipoles then they are tuned to the one frequency and you must plug the correct antenna to the correct SMA. Red connector to Flarm, Blue to ADSB, Black to either.

2.2 Setup Test

Before installing it is best to check your supplied parts are working and to see how you must install them.

1. Plug a USB-C connector into your computer and the WF. Your computer should 'bing' and the green LED should come on. This should happen either way you plug in the USB-C. Ask if this does not happen. We would have tested this so something is wrong.
2. Obtain a power supply of 8 to 36V. Make sure you know the positive and negative of the power supply wires and connect the cable to the WF. Disconnect the USB. Connect the wires briefly to the battery or power supply. Why we say briefly is because it may survive a brief fault. The green LED should flash. If so connect it and the LED should stay on. If not there is a fault. If you smell electrical burning then you have connected it wrong. Send it back to be fixed.
3. Plug your displays into the WF. They should turn on and startup. The SkyViews should 'beep' and change. If you have a SkyView for a tug without a buzzer it will not beep but will talk if a speaker is connected. Others will show lights and change.
4. Now that you know how they must be connected and they work, plan your installation.
5. Pull the wires into place. You can unlock the pins from the SkyView connectors to make the cable go through small holes. Make sure you clip them back into the correct holes, see below.
6. Connect the power supply cables to the glider. Make sure you can turn it off and it is protected by a 1 amp fuse. It can share with other instruments but remember if one blows the fuse you lose them all. Probably best to not share with a radio or another nav instrument.
7. Plug in all connectors and turn on the power. If they are all still working then secure the units with cable ties, Velcro, or even blue tac. Blue tac may sound a bit agricultural but in some locations it makes a good removable mounting, at least temporarily while you become sure of how you want it.

The design intent is that the WAGA Flarm is simply cable-tied to hold it in place and upright. If you want to bolt it then it is best to make a custom bracket to clamp over it. If the bracket goes over the top it must not be conductive, ie plastic not metal or carbon fibre.

The SkyView is best fixed with Velcro so it can be removed. But it can also be double sided taped or even bolted through the back before re-installing the screen and cover. There are very strong rigid versions of Velcro. See below for hints on fixing the antenna. The SkyView is nylon so glue does not bond well. In some ways I like this because if it gets kicked it flies off and does not get damaged. Up to you and your glider. Obviously best if mounted as a recessed instrument.

2.3 Antenna Installation

The rest of the installation is simple and not very critical. But Antennae installation is vital and FLARM® provide a good manual on the subject, download it:

<https://www.flarm.com/wp-content/uploads/2024/04/FTD-041-Application-Note-FLARM-Antenna-Installation.pdf>

Some highlights:

1. Install the WAGA Flarm (WF) with a clear view of the sky above and all round as far as possible. The writing must face up. It contains the GPS antenna which is a modern GNSS device, a uBlox SAM-M8, (designed for automotive installations in the dash). It is best below a non-conductive instrument cover (so not in the full sun) at the top of the instrument housing. It must not be below other devices or below a carbon fibre or metal or other conductive cover or it will be slow to lock on to GPS. It appears to not startup well if on a metal backing so do not attach it to metal cased instruments. The intention is you fasten it securely with cable ties to something at the top of your instrument panel. And preferably out of the sun. It is white to remain cool and the plastic is a type that can take quite a lot of sun. But everything gets extra hot in WA and it may not survive years of abuse. You want to be able to plug in a USB or remove the device to do updates once a year. If you wish to mount it on top of the panel in the sun then we suggest you print it a black cover and bracket of plastic to shade it and provide cooling. It will probably survive being in the sun painted black but that is your risk. It cannot use an external GPS antenna.

2. We elected to provide external antennae for the Flarm and ADSB radios so that you can mount them optimally. This affects the range and blind spots peculiar to your glider/ tug. Take care and check it carefully. Mount them preferably with a least obstructed view all round, vertically, and at least 30cm away from the each other, Flarms and all other antennae. Do not attach them to metal or carbon fibre or other conductive structure.
3. It is best to attach your antenna to a VNA (get a cluey mate to help) and to see how the reception varies as you try different installs. A SWR of 1.5 to 2.0 is best but it may be OK up to 2.5. Being attached to the canopy may change the reception. Try more or less blue tac or different brackets and locations. Do not touch it while measuring.
4. A weakness of the WF (to save >\$500) is it only has one Flarm antennae. This has been tested to work fine in even Pawnees or large Fibreglass gliders like the Stemme. Try to install the Flarm antenna in the best central high location, so it sees all round. You will test it in flight and see the blind spots but it is best to get it in the best spot to start with. The ADSB antenna is less critical.
5. Blue Tac is recommended in preference to double sided tape and other adhesives for fixing the antenna to your canopy. They must not prevent the canopy easily ejecting in an emergency and the blue tac allows them to pull off easily if you eject the canopy. A person in the UK was killed because his installation prevented the canopy ejecting, so it is vital. Blue tac also cleans off the canopy easily and does not damage it but it keeps the antenna in place if you get it right.
6. You can curve your flat antenna a little by carefully warming it with a heat gun and curving it by hand to match the shape you are sticking it to.
7. The advantage of the dipole antenna is they are shorter and can even be installed imbedded in your instrument cover so they fit under a narrow space below the canopy central in the highest spot of the cockpit. Otherwise, the flat antennae are possibly better but essentially the same. Make sure you use plastic or FRP, not conductive brackets. And do not tie the cable to the antenna or the antenna to metal or near metal. Keep it an antenna length away from metal. Mount on Perspex or fabric or a plastic mount should be fine. They must be near vertical.
8. See the FLARM© manual for even more tips. This is the most important part of the install of a Flarm.

2.4 Final Installation

Once you have located the WAGA Flarm (WF) and Antennae:

1. Install the Skyview or other screens. These are probably best installed using Velcro so they can be removed.
2. Work out how to route the cables to the displays and the holes necessary for the cables. You can put the SkyView cables through smaller holes by removing the wires from the connector and then reinserting the wires once through the hole. The displays usually get power from the WF. The space on your panel is going to dictate location most of the time. It is best to be high on the panel so it is in your sight. It can be in a mount to the side. If not certain then you are best to install it temporarily and fly with it for a bit to see how you will use it. The buttons are useful to be able to use but not essential.
3. Connect the power cable to glider power. Be careful of polarity. Connect to the WF. The green light should come on.
4. Plug in the displays and make sure they come on.
5. Screw on the antennae.

2.5 Voice for Tugs and Motorgliders

The SkyView and other Flarm displays have a buzzer which alerts the pilot. This is provided for all aircraft except tugs because they will never hear it. It is effective in a quite glider to alert the pilot to alarms. He then glances at the screen to assist in locating a Threat. However, motorgliders and tugs are best alerted by voice to the headset. Gliders could also use voice which is a provided function by the SkyView but not many other devices. Voice equipment is not supplied with the WAGA Flarm because it is dependent on the aircraft radio and intercom. The following suggestions are given to assist the owner install it if he wishes.

We have tested the following is effective. Most aircraft radios have an auxiliary input or three. However direct coupling of the SkyView to the radio develops noise and the volume needs control.

Therefore, an audio converter like <https://www.altronics.com.au/p/c9620a-2-channel-audio-output-converter/> is a cheap solution and available from Altronics in Perth at about \$22. You also need a 1K quality sealed potentiometer and knob to mount on the instrument panel. And an RCA male plug and some screened cable. You use 1 of the 2 channels on the device. The potentiometer is wired to replace one of the onboard pots. We can supply you with details if you need it.

2.6 Settings

You must set the settings in your Flarm. A tug must be set as a tug and alerts must be turned off or the pilot will get too many alerts. Your aircraft ICAO ID and rego must be entered. And pilot details can be entered for IGC files. All as below.

You must register your Flarm with <http://ddb.glidernet.org/> and <https://www.flarmnet.org/> to make it possible for others and SAR to know who you are. Unfortunately, you need to register with both so all other aircraft can identify you. There are other possible registrations.

WAGA has set your WF with the above and defaults and registered it with glidernet.org and flarmnet.org. There will be other setting you want to set in future.

The WAGA Flarm (WF) and the SkyView or other displays are set in their own way and independently. There are some settings that must be the same like data type and baud rate or the devices will not communicate. Refer to the display manuals/ this manual covers only the WF.

All the settings are given in the manuals from Flarm given in Section 5. These are the required setting commands:

CONFIG COMMANDS	Default	Comment
\$PFLAC,S,ID,<ICAO>	FFFF	#Set ICAO ID from GFA. Often called 24-bit or Hex code. Usually a 6 digit hex code which is assigned to your aircraft.
\$PFLAC,S,NMEAOUT1,0	0	Don't output general strings
\$PFLAC,S,NMEAOUT2,91	91	Output all strings in V9
\$PFLAC,S,ACFT,1	1	#1 is glider, 2 is Tug
\$PFLAC,S,BAUD2,2	2	Flarm is 5 but WF defaults to 2 ie 19,200 is default. Change if your display is set to another baud.
\$PFLAC,S,BAUD1,5	5	Is USB. Default 57,600
\$PFLAC,S,RANGE,<>	65535	Displays aircraft up to Horiz Range, m.
\$PFLAC,S,VRANGE,<>	500	Displays aircraft up to Vertical Range, m.
\$PFLAC,S,PRIV,<>	0	Enable stealth mode
\$PFLAC,S,NOTRACK,<>	0	No ground tracking
\$PFLAC,S,PILOT,<name>		IGC file
\$PFLAC,S,COPILOT,<name>		IGC file
\$PFLAC,S,GliderID,<Rego>		#Rego. IGC file
\$PFLAC,S,GliderType,<>		IGC file
\$PFLAC,S,CompID,<>		IGC file
\$PFLAC,S,CompClass,<>		IGC file

<> replace with the desired text. # must be set correctly and has been by WAGA.

Copy and Paste from here to Putty to WAGA Flarm: As above. Can do all at once. Edit here first.

```
$PFLAC,S,BAUD2,2
$PFLAC,S,NMEAOUT2,91
$PFLAC,S,ID,<>
$PFLAC,S,ACFT,1
$PFLAC,S,GliderID,<>
$PFLAC,S,PILOT,<>
$PFLAC,S,GliderType,<>
$PFLAC,S,COPIL,<>
$PFLAC,S,CompID,<>
$PFLAC,S,CompClass,<>
$PFLAC,S,NMEAOUT1,0
$PFLAC,S,RANGE,65535
$PFLAC,S,VRANGE,500
$PFLAC,S,PRIV,0
$PFLAC,S,NOTRACK,0
```

Most of the above are required for IGC files only or to reduce the number of aircraft displayed and it will work fine as is. Set ID and ACFT and Rego. We have set these for you prior to delivery.

Settings are set using a Display that is capable or a PC. Currently the SkyView cannot set the WF but more complex Flarm compatible displays and glide computers may. Or WF can be set using a PC connected by USB cable and Putty or other serial comm program as follows:

Connect PC to WF. Use Device Manager to see which Com Port it connected to. Open Putty or similar and select the correct Com Port and Baud rate (57,600bps). To see it works type the command \$PFLAS which should list all settings. Then type commands such as given above or the FLARM® manuals. The WF will respond showing what it has done. These settings will be retained until next changed.

You may copy and paste. You can paste multiple commands. Just be sure it is all completely correct before copying. You can get the text from a Flarm-Config tool if you prefer.

2.7 Final Signoff

It is always best to get a third party to check your work. See MOSP 3 Item 18.9. You may install an instrument such as the WAGA Flarm in an Australian registered glider yourself. But you must get it checked for security by an Airworthiness Inspector and signed off in your logbook. You must be sure it is electrically safe, will not interfere in flight, and is crashworthy (ie will not injure you during a crash), and obviously works well. It is useless having a Flarm that does not receive all round you to at least 5 km. You will test this in flight as below. It may be received much further and the modern Atom Flarm that is used in the WF probably has a further range than all other Flarms. They often receive or are received at over 50km but this is not essential for the primary collision avoidance function, it just shows they are sensitive.

2.8 Using the WAGA Flarm

1. Get to know the WF and Display before flight. The WF is simple – make sure it is on. The display is the more complex bit – see separate manual.
2. Don't fiddle with it and keep your eyes out. LOOKOUT is still vital in case you don't get an alert. We designed this unit to be simple and not attract your attention inside. It therefore gives beeps or voice alerts if you connect it to a speaker or the headset. And the SkyView and other displays try to direct your scan in the hazard direction.
3. Flarm has secondary functions of search and rescue tracking on OGN, logbook entry for your airfield, and tracking your mates while flying. But Flarm is not perfect at these functions.
4. When you use the radar display be careful and don't be distracted. If someone is nearby in a thermal or you get an alarm then quick glance and lookout to find him. Be disciplined.
5. If you fly far or get low, OGN will lose you and SAR may be misleading. So rather have a cellphone or satellite tracker. You may crash out of range of OGN and we will find you on the

other SARs but may not on OGN. But if you stay within a few km the OGN/ Flarm should track you and it is useful.

6. It records CARP data for range analysis and IGC files for flight analysis. See Testing for how. You cannot declare tasks but the IGC files are suitable for low security flight tracking.

2.9 Testing

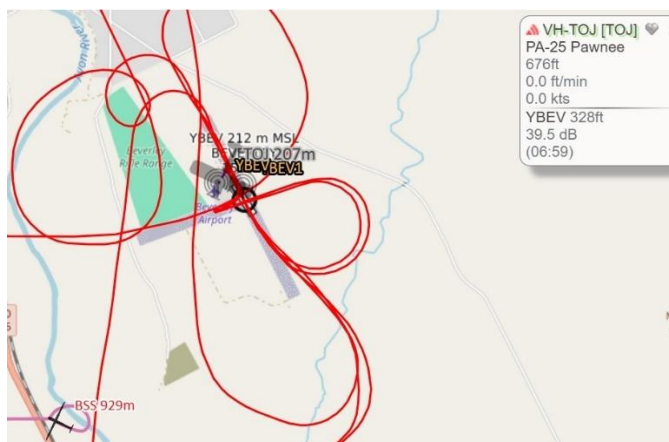
This is the basics of testing. A future GFA manual should discuss it further and in detail as it is a function we need to perform on all Flarms annually.

1. Test on ground before flight that it is received by your airfield OGN with a strength of over 40db when you are as far away as possible (say 0.5km) but in direct line of sight. See below.
2. Go flying with others, the more the better:
 - a. Flarm Transmission strength. You test the transmitter by observing your range and receptions strengths on the OGN receivers, after the flight online, by using Grafana http://ogn-monitoring.org/d/Z3it0u-4z/airport?orgId=1&from=now-1y&to=now&timezone=browser&var-sel_country=AU&var-sel_airport=CUNDERDIN. It is complex and you may get an expert to help. This is a very useful function of the OGN network, ie we can check everyone's Flarm is at least transmitting. For example: [Example Glider data](#). See below for how.
 - b. ADSB-In and Mode-S. You note that you see various GA aircraft that have ADSB or transponders and the range. You should see them even 50km away and usually you see airliners pass over or at Perth. Over time you will see a transponder as a circle of dots, and nearer ADSB aircraft. You could get someone to turn on their SkyEcho and see you receive it a few km away when in flight. This is all there is to test the ADSB-In functions.
 - c. Flarm Reception Strength. You need to have flown with many other gliders at various ranges from you. Download an IGC file from the WAGA Flarm, as below, which contains CARP data which you upload to the Flarm test site. It will tell you how well your Flarm receiver works.
 - d. ie a, b, and c are required to test a Flarm with ADSB-In.

Often the tugs don't show up well on these tests, not because they are not working well, but because they don't fly with others far away at altitude. So be somewhat interpretive of what you get and be patient. However, be sure it works well because there are many Flarms out there that you only see at a km and they are dangerous because they are not going to warn us in time.

2.9.1 Test 1

To do a simple test on the ground; be in sight of an OGN receiver. Open 'glidertracker' on your phone web browser. Find your aircraft on the map. Select it. You will then get a display like:



This also says the last receipt was 06minues59s ago. (I hope YBEV is not at 328ft – that may be m or QNH is standard pressure altitude.)

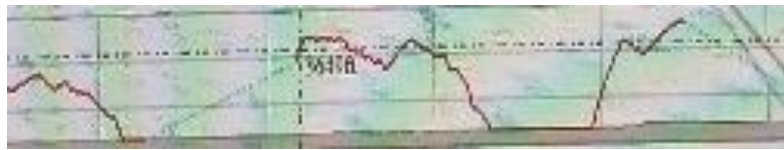
Make sure your rego is in the top right box. Look below the airfield name, 5 lines below, see the reception strength in this example 39.5 db. That is strong. Make sure yours is better than 40db. But it will reduce dramatically if there are trees or further away. When flying it will receive until it is about 5db.



2.9.2 Test 2a

The OGN data is analysed by various systems like Grafana. Too complex to describe here but get someone who knows to help.

However, looking at the barometric trace or the map trace as given in Glidertracker gives you a rough



idea. As alongside, when the trace is green dots it lost the signal. When red it was received. This case shows the Flarm was not on or the GPS had not locked

on for the second flight, then came on and tracked perfectly. During the third flight it worked but lost it for good. The map would show it was within a few km. But when on it worked well and reliably. Often while thermalling one loses a few seconds repeatedly, this is OK if further away but as above it was good.

2.9.3 Test 2c

Reset the CARP data using the command: \$PFLAC,S,RESET sent from a PC to WF. Not required initially but whenever you change your installation, eg move antenna.

Fly with many gliders and they need to go far away from you (>20km) and have reasonable Flarms themselves.

Use SeeYou on a PC connected to the USB on the WF. Note the Com Port from Device Manager. Run File / Connection Wizard. Set Device to Flarm, Interface to Com Port, Parameters to 57600bps, Download... It should show your Atom, SW version and Serial Number. NEXT. Select the flights you want and where to save them. The last flight is probably what you want and contains the CARP data.

Upload it to the FLARM© Range Analysis tool online. It will return an analysis of how well your Flarm is receiving. Fairly obvious. Must be beyond the minimum. And not too lopsided. If lopsided then your antenna is probably badly located. It only shows the horizontal performance and is blind to above and below issues.

Note that you see others when they are above and behind and below. This is subjective but the best we have at the moment.

3 How it works

It is useful to understand a little about how WF works. This is detailed in other manuals but simply:

1. The WAGA Flarm (WF) is programmed by FLARM©. WAGA has no input or control on this. This is best so it works with all other Flarms and is updated by FLARM© to remain compatible. The WF is a fully compatible Flarm.
2. But we have control of many commands and settings for setup and output of data. These were briefly discussed in 2.5 Settings.
3. WAGA can and has improved the programming of the SkyView display. See the manual. We can add to this or improve it. As well as altering settings which change what it displays. For instance, we added navigation to home and we fixed some errors. Most of it was already programmed.
4. WAGA intends to make it easier to update and set the WF. Currently it is setup to work without you needing to do anything before you fly. It even has the database of other aircraft and is registered on the OGN database for you. We realise it is not easy for everyone to do this. We intend to make it easier in future. At the moment ask the few experts to help.
5. All Flarms utilize position and movement information obtained from an integrated GPS/GNSS receiver and optionally from an embedded barometric sensor, (WF has one so it has more accurate altitude). The future flight path of the aircraft is predicted and transmitted over a low-power short-range radio channel as a very short digital message once per second. These messages are received by other Flarm devices within range and then compared with their own predicted flight path. If a collision risk is calculated, Flarm warns the pilot of the chronologically closest collision risk. But

because the antennae may be masked by the glider it cannot be guaranteed or it may disappear for short periods, ie you must still remain aware and lookout.

6. It is useful to be situationally aware and see that others are approaching you or are in the area. Or are behind you. You then know better where to look and how to act. If Flarm can it gives the Rego and you can radio them. They should also be warned of you but if their Flarm receive range is not as good or faulty then they may not.
7. Collision warnings and the three alarm levels are issued depending on the forecast time to impact, not a distance. The first alarm level is usually issued at 18–13 seconds, the second level at 12–9 seconds, and the third level at 8–0 seconds prior to the predicted impact. Each warning lasts as long as the alarm level is applicable. Depending on changes to the forecast, alarm levels might change or disappear. Warnings are highly selective, i.e. they are only issued when there is imminent danger.
8. No Flarm issues resolution advisories, only alarms and situational awareness.
9. ADSB-In and Mode-S transponders work in a similar way but at 1090 MHz. Equipped aircraft transmit a similar signal to Flarm which the WF is equipped to receive and it will display warnings the same as for Flarms. However, Mode-S does not transmit position and so Flarm estimates range and only shows roughly how far away it may be. It still gives alarms the same as for Flarms. But the ADSB symbol has a dark ring around it on the SkyView as it may be a lot faster than a glider and act differently. Likewise tugs have a light ring around them, even if only Flarm equipped.
10. The height above or below is given for the threat aircraft if available. The threat aircraft is the one predicted to collide first. It has a line to it on the SkyView or may be coloured on other displays or show the direction to look. Get to understand your display.
11. The SkyView display can give voice alerts which are useful to tuggies with headsets (or motorgliders). Other displays may give flashing lights. The SkyView has a buzzer to attract attention when required. ie the logic is you lookout until you get a beep, then glance at the display for guidance on where to look. Be disciplined; DO NOT fiddle or stare at the instruments.
12. If set correctly the WF will not warn the tug or the glider on tow = distracting useless warnings. Or about safely thermalling gliders. You can set it to tell you or not about nearby aircraft. Tugs are best set to not tell you or it distracts the tuggie about gliders waiting for launch etc. It will then only Alarm as above, greatly reducing spurious alarms. There are many settings to exclude far away aircraft if you prefer.
13. Obstacle alarms are active but we don't use obstacle databases in Aus and so there will not be any. If you went overseas the WF is worldwide capable and will adjust. You could add an obstacle database to the SD card and it should work. Untested. You may need to change your Flarm antenna.

4 Support

4.1 Software Updates

To update your WF requires you download the latest version for the Atom Device Type from the FLARM© update webpage. We assume you know enough or get help.

Connect the WF's USB-C to a PC USB port:

1. You need to have Python and serialutil loaded on your PC as per separate instructions.
2. Use the PC Device Manager to look up the COM port number for the device
3. Use this in the following command to load the firmware in CMD prompt:

```
python -m serialutil --port COMxx update_fw_recovery c:\atom3.fw
```

4. Where: xx is the COM port number and atom3.fw is the firmware update file.

We intend to make this simpler before you need to do this. It should not be needed until May 2026.

4.2 Technical Assistance

The WAGA Flarm (WF) hardware is provided without warranty or guarantee of continuing support. However, WAGA designed and developed it and we intend to share the data so that the devices can be serviced probably quicker and cheaper than foreign devices. We retain some spares.

WAGA members using a WF should contact their Club designated 'WAGA Flarm and SWD Super User' for help and advice. Please report defects and issues so we can address them.

5 Further information

There are abundant configuration settings possible and a few needed in the WF. Most of the needed settings can be done from a Flarm compatible display but probably more easily using a PC and Putty. The commands are all defined in:

FLARM FTD-012 : Data port interface control document

FLARM FTD-014 : Flarm configuration specification

FLARM FTD-041: Application Note FLARM Antenna Installation

Available from the FLARM© downloads webpages. There are also many more useful manuals although most people will not need more than the Antenna manual.

LXNav produce a simple \$200 device to enable any Flarm to be connected to your Smartphone by WiFi, called a LX FlarmLink. It is available in Australia at <https://store.swiftavionics.com.au/lx-flarm-link/>. A couple of people have one and it could be shared/ borrowed. It allows easy setting and probably updates. We will test further and maybe each club should have one. We may be able to get the SkyViews to do this.



6 Acknowledgements

Kevin Wilson of BSS is greatly acknowledged for his months of work in programming the SkView display. It was meant to be good enough to go but it was found lacking and in error on important aspects. Kevin worked it out and has it working well. We are likely to continue to develop it and it is easy enough to upgrade the software and get the latest version in future.

Kevin and Mike Eales (BSS and GCWA) provided input on the electronic design and programming of the WAGA Flarm.

The GCWA tug pilots tested and commented on the prototype and resulted in the SkyView being improved.

Rob Hanbury (GCWA) designed and developed the concept and the board for the WF.

We acknowledge and appreciate the support that FLARM© provided from the initial concept stage right through development and provision of the latest firmware. They will continue to support the firmware and provide many manuals and the update downloads on their web pages.

Appendix A: WAGA Flarm Cables and Connections

1 Connector Options

1. The standard configuration is; 12V from the glider battery to the Data Port, a display is connected to one or two of the pigtails and/ or the Data Port, two antennae are always connected.
2. SkyViews to the pigtail connectors plus power in from the glider battery to the Data Port. Power in from USB (5V) for regular use is not recommended as the connector is not durable.
3. SkyView to a pigtail plus power in and a data connection to a Flarm compatible display or glide computer via the Data Port. The WF will provide 3.3V to 3.4V to the displays.
4. Data and power via the USB, not recommended in flight as the connector is not durable. It is intended for occasional use on the ground to connect to a computer for configuration.
5. The pigtails can be connected to 3.3V or 5V dependent what your display requires. It is usually 3.3V and is supplied at such.

The same data is output at the same Baud rate and protocol to the pigtails and the Data Port. The settings set both and both are Port 2.

The USB is Port 1 and is the primary (update) port but otherwise port 1 & 2 can both receive the same commands and output the same data. But they can be set differently using items NMEAOUT1 and NMEAOUT2 work like NMEAOUT but configure a specific data port (No. 1 or No. 2). Likewise BAUD.

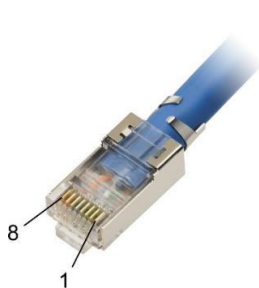
2 Cables

2.1 Data Port (RJ45): Power and Data Connector



Connecting power to an incorrect pin will destroy components and make the device unusable.

The 8-pin RJ45-socket (8P8C) is in accordance with IGC GNSS FR specifications, except for pin 6.



1. GND
2. GND
3. RX, Flarm receives (RS-232)
4. TX, Flarm sends (RS-232)
5. GND
6. All Flarm supply +3.4 V DC for a display
7. +12 to +32 V DC power supply
8. +12 to +32 V DC power supply

Note: The pin numbering above follows international standards. The pin numbering in IGC documentation and early PowerFlarm manuals was reversed; however, the actual pin assignment is the same as in the older manuals. The pin numbering in different display documentation is mixed. See the picture above for correct pin assignments.

Note: Pins 7 and 8 of the RJ45 connector are internally connected. So are the Ground pins 1, 2, and 5 of the RJ45. You can insert a 6 or 8 pin connector, an 8 pin is best.

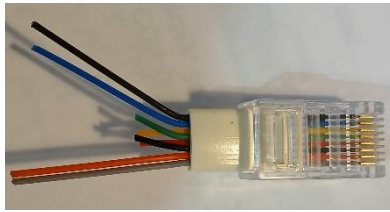
Note: Do not connect more than one external device to the RJ45 port.

Note: Maximum current from pin 3: 200 mA @ 3.3 V DC.

Note: The wire colours may vary from that shown although we tried to use Black for ground, Red for power of any voltage, yellow & green for TX and RX (but these are dependent on which end of the cable). But as below we sometimes had to use other colours.

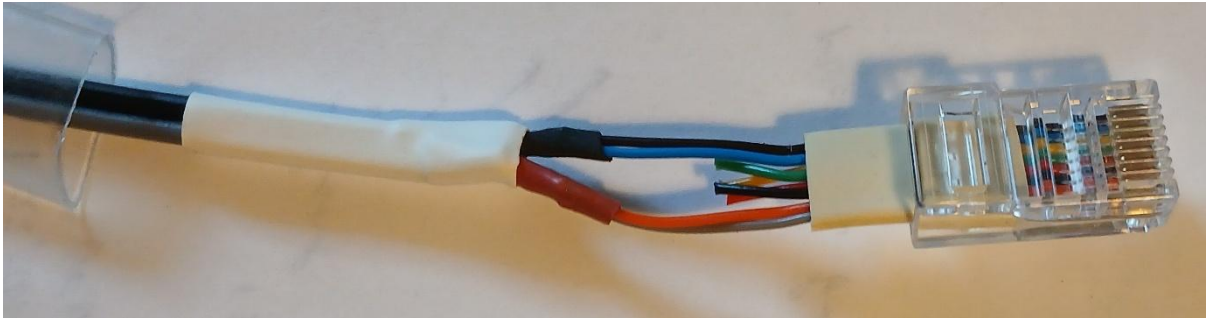
2.2 Power to Data Port Cable

This cable is in the WAGA Flarm kit if you ordered SkyView displays.



8-core modular flat telephone cable to 8P8C plug. Left 2 wires are ground. Right 2 wires are +8 to 36V, usually 12V glider battery. Other 4 wires are trimmed.

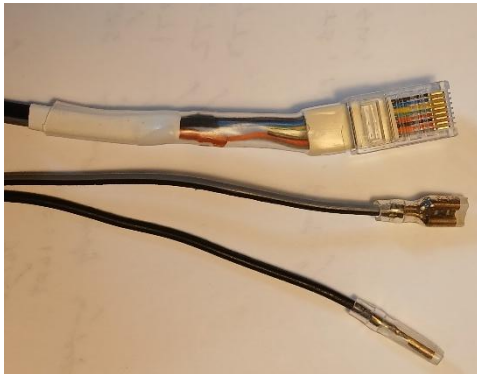
Connect 1 Amp rated cable to plug as shown. Be careful of orientation, shorts and durability.



Final cable showing polarity marked cable, heat-shrink insulation, and mechanical security.

Data is not connected. Only power in is connected.

If power is on then the green LED on the WF will light. Your displays should turn on. The SkyViews, being ePaper displays, remain on the display at turn off and can be rather hard to see if on or off.



2.3 Power and Data to Data Port Cable



Similar to 2.2 above but connects to displays as well using the Flarm standard 6-pin plug and to power in from the glider. This is not supplied by WAGA as you probably have one for your display. But this is how to make one:

1. 8-core modular flat telephone cable to 8P8C plug. As shown above:

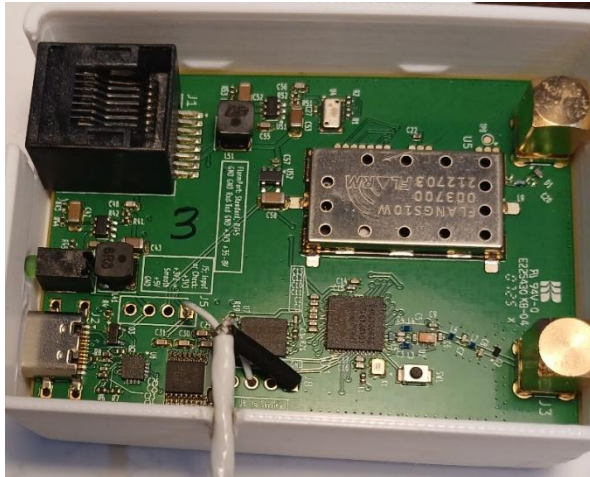
No.	Wire Colour above	Function
1&2	Brown & Blue	Left 2 wires are power ground
3	Yellow	Data in (RX)
4	Green	Data out (TX)
5	Red	Data ground
6	Black	3.3V out
7&8	Orange & Grey	Right 2 wires are 8 to 36V, usually 12V glider battery.

2. The 6-pin end of the cable is trimmed to fit a 6P6C modular flat telephone plug. Cut the sheath edges at the entry to the plug, leave the Brown and Grey wires folded back. The central 6 wires are inserted in the same orientation, as shown, as the 8-pin plug. The sheath on both sides is left long to insert into the plug and provide mechanical strength.
3. Solder 1 Amp rated cable to the Brown and Grey wires as shown. Be careful of orientation, shorts and durability.
4. The modular plugs plug into the WF and the Flarm compatible display or glide computer. The power wire is connected to a fused or trip-switch power supply in the glider. Fuse of 1A is sufficient. This can have its own switch but in some ways it is best to come on by default with all instruments so the Flarm is not left off by mistake. The advantage of a switch for some instruments is a switch can isolate a fault and allow other instruments to stay on.
5. A similar cable supplied for other Flarms and displays should be compatible and be useable as is. Best to test power polarity to be sure.

2.4 Pigtails to SkyView

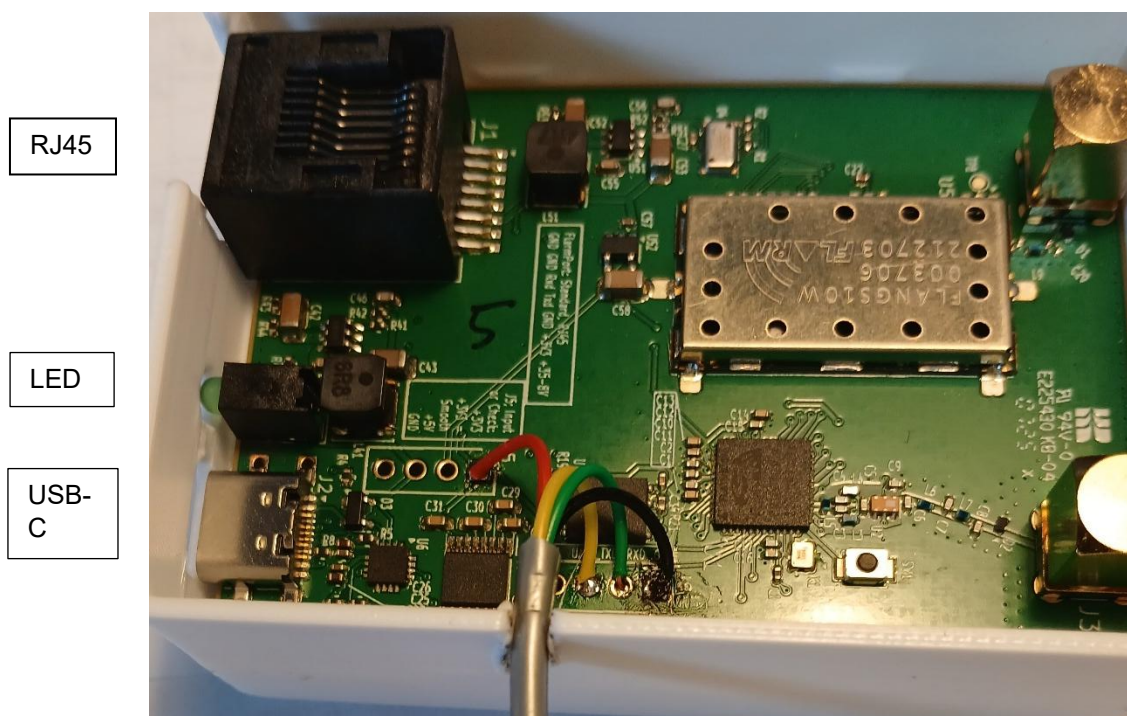
The SkyViews are different to other Flarm displays and require TTL signals, not RS232. They will not work directly on the RJ45 socket. However, they share the same output and receive the same baud rate and all signals. The baud rates must be set the same on all the display devices and the WF (they are usually flexible to use almost any baud rate $\geq 19,200$ bps).

If you ordered a SkyView then your WF will come with one or two pigtails for a front and a rear SkyView. You may change it as follows.

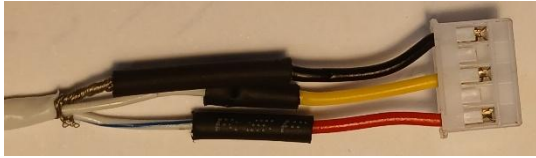


2-core and shield tefsel cable connected to board. Board is labelled; J5 is power = 3.3V and 5V, J8 is data and can provide 5V if desired. GND is on either. Smooth 3.3V is not required for displays.

Below shows 4 core telephone flat cable connected to J5 to TX, RX, GND and on power J8 to 3.3V. (Note that yellow is not the same as the yellow in the RJ45 cables.)



You may parallel two pigtails to the same connections but rather only connect one to the RX. If you have 2 pigtails neither have to be connected. You may also cut them off if no longer required.

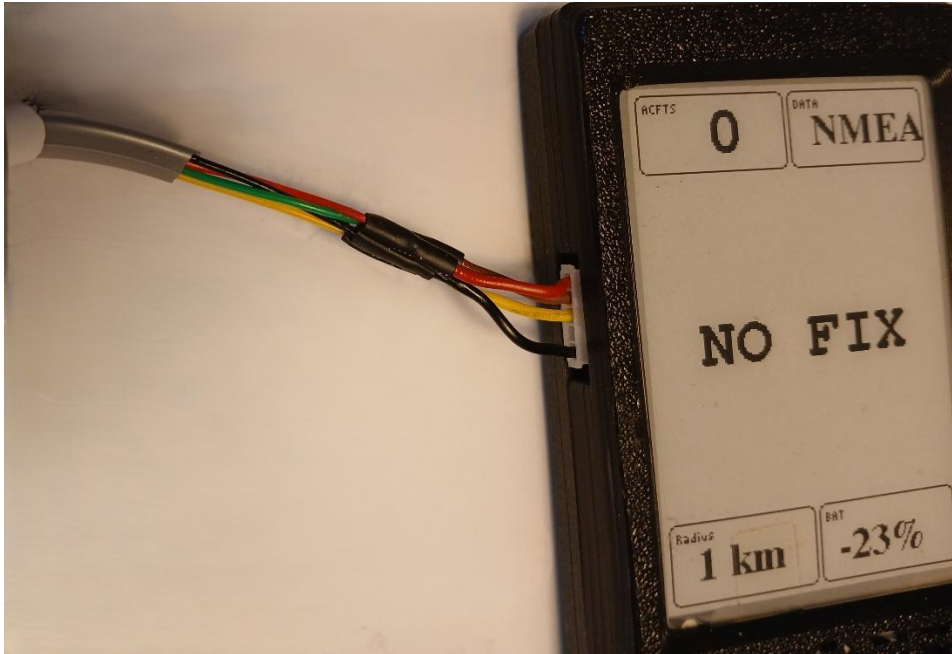


Remove all wires from the holder for the SkyView connector, use a pin or blade to unlatch. Cut wires to length. Connect by soldering and heat-shrink to the pigtail. Ensure correct connection. Slip on a piece of outer heat-shrink long enough to cover all the wires.

As above; Black is Ground, yellow is TX from Flarm to RX on SkyView, and red is +3.3V.

Note that the output is called +3.3V but usually outputs 3.4V. The smoothed supply will be 3.3V. The 5V supply can be as low as 4.8V to 5.1V.

Insert the cable through a small hole in the instrument panel before inserting the wires in the connector. Plug the wires and terminals back into the correct slots on the connector, in the correct orientation. They click in easily and latch.



If the green was connected it would go between the yellow and red, TX from SkyView to RX on Flarm. But as shown here the green wire is connected to a brown wire (as supplied with SkyView).

Slide up the heat-shrink. Rather bend the cable to suit the installation then heat the heat-shrink to form a neat bend.